

SAFWAN YASFEE FAHIM

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[GitHub](#) • [LinkedIn](#)

Education

American International University-Bangladesh
Bachelor of Science in Computer Science

January 2023 – Present

Safiuddin Sarker Academy & College, Gazipur
Higher Secondary Certificate (HSC) in Science
GPA: 5.0

2021

Profile

An undergraduate in Computer Science and Engineering, focusing in Information Systems, with practical experience in database design, systems analysis, and business process automation acquired through academic and independent projects.

Research Experience

Research Assistant-Energy and Technology Research Division

- Developing a continuous-time ODE Transformer framework to address asynchronous SCADA telemetry challenges in renewable energy grids
- Combining Neural ODEs and Transformer architectures to model irregularly sampled grid telemetry data

AI-Based Student Performance Prediction and Academic Risk Analysis System

- Designed an academic early-warning system using the CRISP-DM framework, comparing Decision Tree, Random Forest, KNN, Logistic Regression, XGBoost, and CatBoost classifiers
- Domain-specific features (academic consistency indices, burnout/effort ratios, attendance risk flags) and cost sensitive weighting/threshold optimization implemented

Projects

Dhaka Road & Draining Issue Reporting System

- Developed frontend with Next.js 14, Axios (JWT auth), Zod validation, React Hook Form, hooked to certain backend API endpoints
- Utilized principles of information systems design to organize a real-world civic issue-reporting process

Analysis of Factors Influencing Video Games Success on IMDb

- Automated data ingestion and ETL: Created an automated web-scraping pipeline in R (rvest, stringr) to extract, clean and structure unstructured metadata from 25+ top-tier titles on IMDb.
- Exploratory Data Analysis (EDA): Developed text-cleaning routines using regular expressions (Regex) and conducted non-parametric EDA (ggplot2) to evaluate skewed distribution patterns across audience age ratings.

- **Statistical Modeling & Validation:** Built a Simple Linear Regression model (lm) to compare community engagement with public rating scores. The model yielded a $R^2 = 0.65$ ($p < 0.05$).
- **Critical Evaluation & Reporting:** Identified key dataset constraints (survivorship bias in top-tier samples) and wrote an executive-ready technical report outlining actionable future expansions, including Natural Language Processing (NLP) sentiment analysis.

Technical Skills

Information System: System Analysis & Design (SDLC), Business Process Modeling (BPMN/UML/ERD), ERP/CRM/SCM concepts, Requirements Gathering

Database: SQL, Data Modeling, Power BI, Oracle, Microsoft SQL Server, MySQL, PostgreSQL, MongoDB, Transformers, Neural ODEs

Programming: Python, JavaScript (ES6), Typescript, C++, java, C#

Tools & Frameworks: Git, GitHub, VS Code, Microsoft Visual Studio, GitHub Copilot, Postman, Next.js, React, Zod, Axios, OpenGL

Certifications & Training

IT Essentials | *CISCO Networking Academy* | 70 hours | Jan - May 2023

Next Level AI-Driven Software Engineering Bootcamp | *Programming Hero Level 2* | 6 Months | May 2026 – Present